

The empirical recurrence for OEIS A251006: recovering a conjecture that is not written down

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Abstract

OEIS A251006 records “Empirical recurrence of order 87”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 243$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 87, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 87 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A251006 is “Number of $(n+1) \times (4+1)$ 0..2 arrays with no 2×2 subblock having its maximum diagonal element less than the absolute difference of its antidiagonal elements.”. It has offset 1 and begins

31456, 4434257, 623999793, 87861924805, 12371981931569, 1742128826665123, 245313779556192507, 3454330661

The entry states:

Empirical recurrence of order 87 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 12:58:39, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 243$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 243$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 486$ exact terms of the model returns order 87 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{87} c_i a(n-i),$$

c_1	243	c_{23}	−350416340703436310165956982	c_{45}	−9598730701328796719
c_2	−20493	c_{24}	1310053014008769100611217140	c_{46}	95147563120693000795
c_3	1137320	c_{25}	−4069404609077554356080488904	c_{47}	−6764873493034973828
c_4	−48968370	c_{26}	9553096110498373060623750168	c_{48}	11015145502469979239
c_5	1660074126	c_{27}	−11390048466311802311290569758	c_{49}	65936287947907648082
c_6	−44634837818	c_{28}	−31247919197470577449527684590	c_{50}	−14403210142600165189
c_7	967746677672	c_{29}	261318622987444316954896951975	c_{51}	200328497373182890299
c_8	−17068124680628	c_{30}	−1032420856145510224087880547173	c_{52}	−21746737653130442457
c_9	243689314141870	c_{31}	2978242929861329612880677440996	c_{53}	191023979529407639163
c_{10}	−2771271784376738	c_{32}	−6834017778181258768810376827115	c_{54}	−13133290144566069777
c_{11}	24244713468588507	c_{33}	12711258476587812926162830736737	c_{55}	59165700478091969081
c_{12}	−146741653931132321	c_{34}	−18557557689923271217607264914481	c_{56}	2978228756926141798
c_{13}	290165718947140921	c_{35}	17681774295226746906280242801957	c_{57}	−4010565310845367397
c_{14}	6772301199597629044	c_{36}	4535426194989267465243562688612	c_{58}	49704045171127221780
c_{15}	−120128686970779626924	c_{37}	−74547830576501127926017986796070	c_{59}	−3985948792481019808
c_{16}	1290337980725726668601	c_{38}	235915091706266418767778764894910	c_{60}	22593074034727099416
c_{17}	−10997095841855253481106	c_{39}	−558303972809578921355842829140242	c_{61}	−7363003674637401418
c_{18}	80531766804597322929398	c_{40}	1149235873598640209626628103083624	c_{62}	−172512012230111394
c_{19}	−525717119916835093119093	c_{41}	−2152009249958486012157521262691128	c_{63}	4992606532325458086
c_{20}	3110972934649033445604839	c_{42}	3693465780073345062916544686127700	c_{64}	−473013558067733674
c_{21}	−16735849846923323800937993	c_{43}	−5752653298321353700715534979870528	c_{65}	3182756061292832761
c_{22}	81244203916263823067469744	c_{44}	7981639485078402516851358413364395	c_{66}	−168912174536011328

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 87, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $87 + 4$ terms removed returns order 87 as well, so $\deg q_{\min} = 87 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 11 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $87 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 87 that this sequence satisfies — and that family turns out to have exactly one member.

References

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